

Q1)

1. Write a program to create a 2D NumPy array of shape (3, 3) with all elements initialized to 0.
2. Write a program to create a 1D array of numbers from 0 to 19. Extract the elements from the 9th to the 11th index.
3. Write a program to print the shape, size, data type, and number of dimensions of the array.
4. Write a program to compute the mean, median, average of a NumPy array.
5. Write a program to create a 2D NumPy array of shape (3, 3) with all elements initialized to 1.
6. Write a function that divides two numbers and handle the case where the divisor is zero.
7. Write a function that asserts the divisor is zero.
8. Write a function that asserts the given number is positive.
9. Write a program that reads a list of numbers and asserts that all numbers are positive.
10. Write a function that raises this exception if a negative number is passed.
11. Write a program to raise a user defined exception to check if age is less than 18.
12. Write a Python program to read an entire text file.
13. Write a Python program to read first n lines of a file.
14. Write a Python program to read data from a Specific Position.
15. Write a Python program to inserting Data at a Specific Position
16. Write a Python GUI program to create a label and change the label font style (font name, bold, size) using tkinter module.
17. Write a Python GUI program to import Tkinter package and create a window. Set its title and add a label to the window.
18. Write a Python program to append text to a file and display the text.
19. Write a Python program to resetting the File Pointer to read again.
20. Write a Python program to reading a file in backward direction.

Q2)

1. Write a Python program to reading a file in backward direction.
2. Write a Python program to copy content of one file to another file.
3. Write a Python program to capitalize the content while writing the content into the second file.
4. Write a program to define custom exception as SalaryNotInRangeError. (Salary must be in between 10000 to 50000)
5. Python program for FiveDivison Error Exception.
6. Create two lists, one representing subject names and the other representing marks obtained in those subjects. Display the data in a pie chart and bar chart.
7. Generate a random array of 50 integers and display them using a line chart, scatter plot, histogram and box plot. Apply appropriate color, labels and styling options.(Apply title, legend, xlabel, ylabel)
8. Create a Tkinter window with multiple labels displaying different text messages and styled with different fonts and colors.
9. Create a Tkinter window with a button. When the button is clicked, display a message in a label.
10. Create a Tkinter window with a button. When the button is clicked, display a message using messagebox.
11. Create a Tkinter window with two frames. The first frame should contain a label and the second frame should contain two buttons side by side.
12. Create a Tkinter window with a frame containing three buttons. Arrange the buttons vertically within the frame.
13. Create a Tkinter window with multiple labels and place them using grid.
14. Write a Python program to create a Tkinter-based login form with input fields for userid and password.
15. Write a python program for addition of 2 numbers using GUI (Use tkinter – Label, Entry, Button)
16. Write a Python program to compute the number of characters, words and lines in a file.

17. Write a Python program to read student information and write it in "student.txt".
18. Write a function that reads a file and converts its content to an integer. (hint-Handle FileNotFoundError and ValueError.)
19. Write a function to read a file and print its contents. (hint -Use try-except-else to handle file not found errors.)
20. Use sales.csv which contains sales data including transaction ID, product, amount, and date
 - a. Read data from a CSV file into a DataFrame.
 - b. Display first 5 records and last 5 records.
 - c. Check Null and duplicate values.

Use students.csv dataset which contains student names, ages, grades, and cities. Write a Python program to

- a. Read data from a CSV file into a DataFrame.
 - b. Add a new column GPA and remove an existing column cities.
 - c. Select specific columns and rows using labels and indices.
 - d. Check Null and duplicate values.
 - e. Display statistical information
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2. Use sales.csv which contains sales data including transaction ID, product, amount, and date
 - a. Read data from a CSV file into a DataFrame.
 - b. Display first 5 records and last 5 records.
 - c. Display first 3 records and last 7 records.
 - d. Select specific columns and rows using labels and indices.
 - e. Check Null and duplicate values.
 - f. Display statistical information.

1. Write a function that converts a string to an integer and handle cases where the string is not a valid number.
2. Write a Python program to append text to a file and display the text.
3. Write a Python program to resetting the File Pointer to read again.
4. Write Python program to copy odd lines of one file to another file in Python.
5. Write a Python program to read last n lines of a file.
6. Write a Python program to count the number of lines in a text file.
7. Write a Python program to print each line of a file in reverse order.

8. Write a Python Program to perform the following:

Write Operation:

The `write_to_file()` function opens a file in 'write' mode ('w'), which creates the file if it doesn't exist, or truncates it (empties it) if it does exist.

The content is written into the file.

Read Operation:

The `read_from_file()` function opens the file in 'read' mode ('r') and reads the content.

If the file does not exist, a `FileNotFoundError` is caught and a message is returned

9. Write a Python program to print each line of a file in reverse order.